

Homework 01 Solutions

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Course: 10-783 A
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Task 1.1

We have that X is a zero-centered Gaussian since e^{-t} is a deterministic function of time. Then, by Ito isometry,

$$\text{Var}(X) = \mathbf{E} \left[\left(\int_0^2 e^{-t} dW_t \right)^2 \right] = \int_0^2 e^{-2t} dt = \frac{1 - e^{-4}}{4}.$$

Thus, $X \sim \mathcal{N}(0, (1 - e^{-4})/4)$.

Task 1.2

Let $f(t)$ be the piecewise function

$$f(t) = \begin{cases} 1 & 0 \leq t \leq 1/2 \\ 2 & 1/2 \leq t \leq 1 \end{cases}.$$

Because $f(t)$ is time-dependent, X is a zero-centered Gaussian with variance

$$\text{Var}(X) = \mathbf{E} \left[\left(\int_0^1 f(t) dW_t \right)^2 \right] = \int_0^1 f(t)^2 dt = 2.5.$$

Thus, $X \sim \mathcal{N}(0, 2.5)$.

Task 1.3

We have that

$$\begin{aligned} \text{Var}(X) &= \mathbf{E} \left[\left(\int_0^1 t dW_t \right)^2 \right] = \int_0^1 t^2 dW_t = 1/3 \\ \text{Var}(Y) &= \mathbf{E} \left[\left(\int_0^1 (1-t) dW_t \right)^2 \right] = \int_0^1 (1-t)^2 dW_t = 1/3 \\ \text{Cov}(X, Y) &= \int_0^1 t(1-t) dt = 1/6. \end{aligned}$$

Thus, X and Y are not independent because they have a non-zero covariance.

Task 1.4

Task 1.10

(a) True. We have that

$$\int_0^1 f(t) dW_t = \lim_{|\Pi| \rightarrow 0} \sum_{i=1}^{n-1} f(t_i)(W_{t_{i+1}} - W_{t_i}),$$

where each interval $W_{t_{i+1}} - W_{t_i} \sim \mathcal{N}(0, \Delta t_i)$ and $W_{t_{i+1}} - W_{t_i} \perp W_{t_{j+1}} - W_{t_j}$ for $i \neq j$. Thus, for any partition Π , $\int_0^1 f(t) dW_t$ is a weighted sum of independent Gaussians, so it must be normally distributed.

Task 3

	t	E[W_n(t)]	theory=0	Var[W_n(t)]	theory=t
1					
2					
3	0.100	-0.00473	0.000	0.10167	0.100
4	0.250	-0.00961	0.000	0.25102	0.250
5	0.500	-0.01644	0.000	0.50755	0.500
6	0.750	-0.02026	0.000	0.77095	0.750
7	1.000	-0.02012	0.000	1.02874	1.000

(2) See Figure 1.

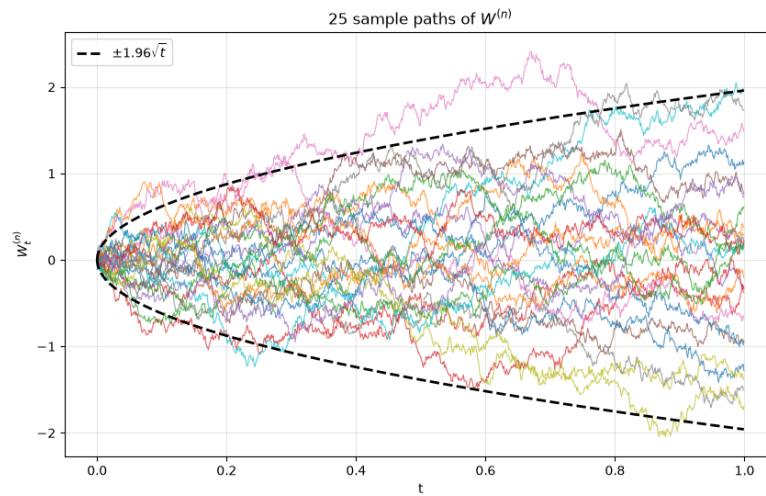


Figure 1: 25 generated Wiener processes with 95% envelope

(3) See Figure 2.

1		empirical	theoretical
2			
3	mean	-0.00479	0.00000
4	variance	0.39814	0.40000

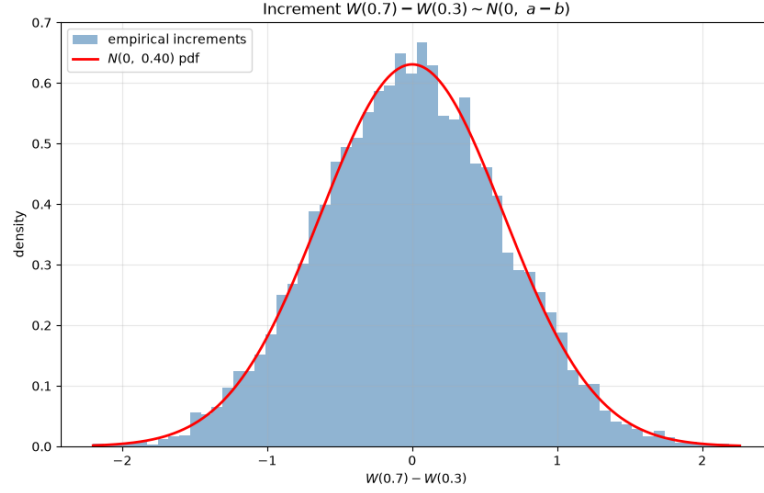


Figure 2: Empirical distribution of Wiener process increments

Task 4.1

For I_1 , the integrand is a deterministic function of time, so it must be Gaussian distributed with mean 0. The variance is given by Ito isometry

$$\text{Var}(I_1) = \mathbf{E} \left[\left(\int_0^1 t \, dW_t \right)^2 \right] = \int_0^1 t^2 \, dt = 1/3,$$

giving that $I_1 \sim \mathcal{N}(0, 1/3)$, as desired.

Task 4.2

See Figure 3. We use $m = 10,000$ Wiener trajectories and $n = 5000$ subintervals of $[0, 1]$, evaluating the integrand at the left endpoint. The empirical distribution is close to the theoretical $\mathcal{N}(0, 1/3)$ distribution.

Task 4.3

See Figure 4. For each $n \in \{10, 20, 50, 100, 200, 500, 1000, 2000, 5000\}$, we calculate all three tagged sums on the same $m = 10,000$ trajectories and take the maximum absolute differences across trajectories. We have that

$$\begin{aligned} S_n(t, \text{mid}) - S_n(t, \text{left}) &= \frac{W_1}{2n}, \\ S_n(t, \text{right}) - S_n(t, \text{left}) &= \frac{W_1}{n}. \end{aligned}$$

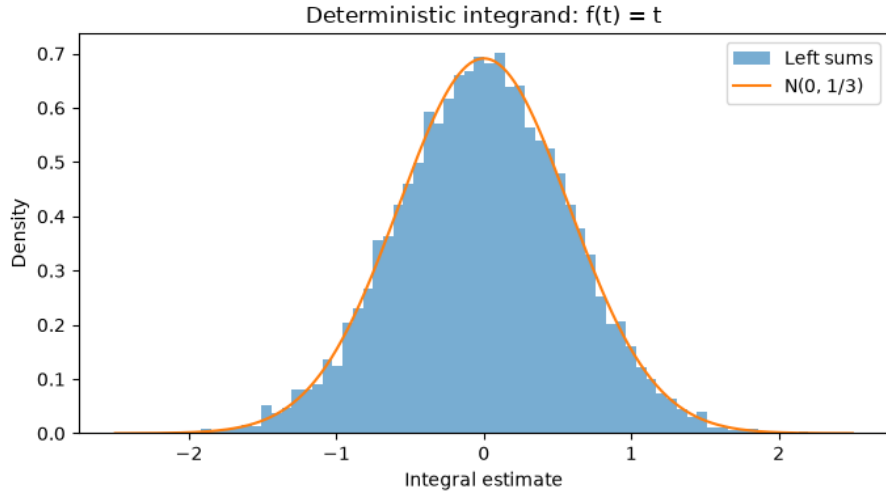


Figure 3: Empirical distribution of left tagged sums for $\int_0^1 t \, dW_t$ with the theoretical Gaussian density

Thus, both differences converge to 0, so the tag choice does not affect the limiting integral.

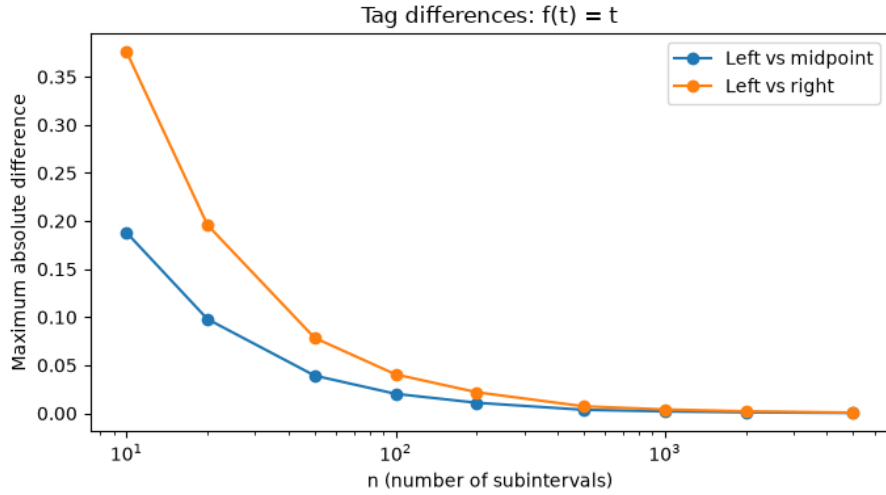


Figure 4: Maximum absolute left-midpoint and left-right tag differences for the deterministic integrand $f(t) = t$

Task 4.4

See Figure 5. We again use $m = 10,000$ trajectories and $n = 5000$ subintervals, now with $f(t) = W_t$ and left endpoint tags. By Ito's formula,

$$I_2 = \int_0^1 W_t \, dW_t = \frac{W_1^2 - 1}{2}.$$

Thus, the limiting distribution is a shifted and scaled χ_1^2 distribution, rather than a Gaussian, consistent with the histogram. We use dW_t as specified in the definition of the stochastic sums and the caption of Figure 3 in the handout.

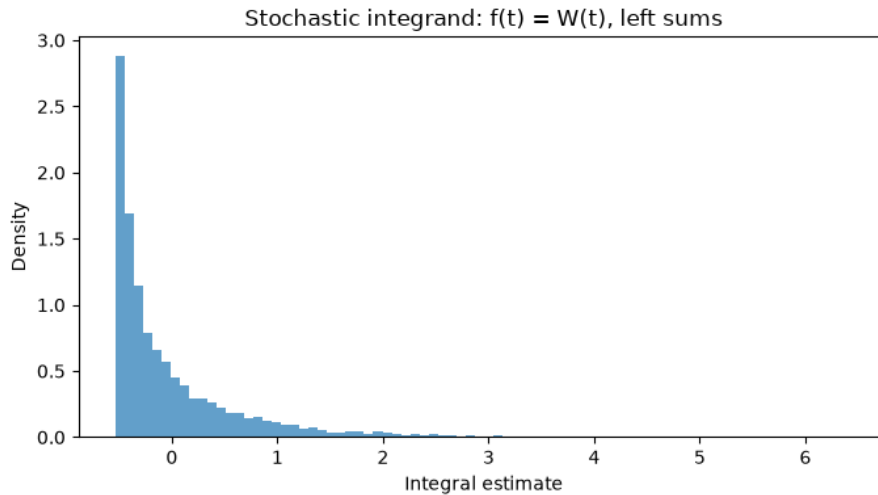


Figure 5: Empirical distribution of left tagged sums for $\int_0^1 W_t dW_t$

Task 4.5

See Figure 6. We repeat Task 4.3 with $f(t) = W_t$, using the same values of n and m . The midpoint tags use the Brownian values at the midpoint times, obtained by simulating on a grid with $2n$ steps. The limits are

$$S_n(W, \text{left}) \rightarrow \frac{W_1^2 - 1}{2},$$

$$S_n(W, \text{mid}) \rightarrow \frac{W_1^2}{2},$$

$$S_n(W, \text{right}) \rightarrow \frac{W_1^2 + 1}{2}.$$

The left-midpoint and left-right differences approach $1/2$ and 1 , respectively. Thus, the tag choice matters for this stochastic integrand.

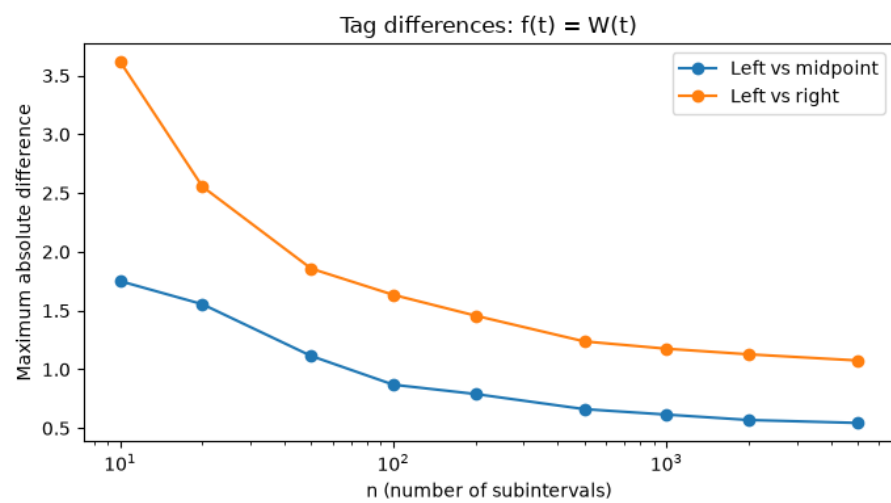


Figure 6: Maximum absolute left–midpoint and left–right tag differences for the stochastic integrand $f(t) = W_t$

Collaboration Questions

- (1) (a) Did you receive any help whatsoever from anyone in solving this assignment? Yes.
(b) If you answered 'yes', give full details. OpenAI Codex helped implement and execute the notebook for Task 4, generate the figures, and write the corresponding explanations in this document.
- (2) (a) Did you give any help whatsoever to anyone in solving this assignment? No.
(b) If you answered 'yes', give full details. N/A.
- (3) (a) Did you find or come across code that implements any part of this assignment? Yes.
(b) If you answered 'yes', give full details. OpenAI Codex generated the initial implementation in `hw2/coding/Stochastic_Integrals.ipynb` using the provided Wiener process notebook as a reference. I edited that notebook, and Codex added the figure export calls.